

03_analysis

Helix OTA — Analysis of Addition #3 (initial_research_03/ helix_ota_big/)

Field	Value
Revision	1
Created	2026-06-08
Last modified	2026-06-08
Status	active
Status summary	Exhaustive requirement-by-requirement analysis of the large generated research/design corpus in additions/initial_research_03/helix_ota_big/ (26 markdown files, ~36k lines). Maps every requirement to the current canonical spec corpus (docs/research/main_specs/) and the ACTUALLY-BUILT server (server/). Reconciles the addition's own CODE_REVIEW_AND_GAP_ANALYSIS.md against the real build, flags conflicts with locked decisions, and catalogues future-phase docs to their versioned dirs. This is Rev-1 coverage of addition #3 — the prior additions_synthesis.md (Rev 1) did NOT cover it.
Issues	The addition is a parallel, self-consistent-but-divergent generation: different module path (vasic-digital/helix-ota-server vs built HelixDevelopment/helix_ota), different HTTP framework (chi vs LOCKED Gin), different crypto (RSA-4096/RSA-PSS vs built ed25519), different domain vocabulary (updates+rollouts vs canonical

Field	Value
	releases+deployments), and mTLS device auth vs built JWT-bearer. Its internal CODE_REVIEW audits the addition against ITSELF, not against the built system — its findings are re-scored here against the real build.
Issues summary	Treated as authoritative INPUT (D5) but NOT as canonical: where it conflicts with locked decisions or the built server, the locked/built side wins; reusable material is folded forward.
Fixed	Addition #3 fully processed (priority docs read in full / deep; future-phase docs catalogued).
Fixed summary	All 26 .md files inventoried; binaries/exports (.pdf/.docx/.html/.tar/.zip) intentionally skipped as duplicates.
Continuation	Fold accepted MVP gaps (user/device-group/api-key CRUD, telemetry-event endpoint, mandatory-update behavior, anti-rollback check, tamper-event, one-active-deployment-per-group constraint) into endpoints.md/schema.md/server/. Route future-phase material into the versioned dirs listed in §6. Re-run when addition #4 lands.
Owner	Helix OTA spec-analysis
Related	<u>additions_synthesis.md; ../../1.0.0-mvp/api/endpoints.md; ../../1.0.0-mvp/database/schema.md; ../../1.0.0-mvp/server/architecture.md; ../../00-master/2026-06-07-helix-ota-design.md</u>

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§1. method and evidence base

Three lenses per the operator mandate (D5; Constitution §11.4.6 no-guessing, §11.4 anti-bluff): **extract** reusable material, **reconcile** against the canonical corpus + built code, **route** each element to its destination. Evidence cited by file. Claims that could not be verified against the built code or live sources are marked **UNVERIFIED**.

Built-server ground truth (read this revision): `server/internal/api/`
`server.go` (Gin router + routes), `wire.go` (wire types), `store/store.go`
(Repository + domain structs), `handlers_client.go`, `handlers_deployment.go`.
Key facts:

- Module: `github.com/HelixDevelopment/helix_ota/server`; depends on `HelixDevelopment/ota-protocol`, `ota-artifact-validator`, `ota-telemetry-schema`, `ota-rollout-engine`.
- HTTP framework: **Gin** (`gin-gonic`) — matches LOCKED D6.
- Persistence: **in-memory** `MemoryRepository` (`store/memory.go`); `pgx/` `PostgreSQL` is a documented future target (`store.go` header, `architecture.md` §4). No DB built yet.
- Domain vocabulary: **devices** → **artifacts** → **releases** → **deployments** → **telemetry**. Deployment strategy = `all-targets` only; staged rollout deferred to 1.0.1 (routes the cohort check through `ota-rollout-engine.InCohort` so the staged engine drops in).
- Artifact signing: **ed25519 detached signature + SHA-256** (`server.go` `ed25519.PublicKey`, `pubKey`), not RSA-4096.
- Auth: **JWT bearer** (`TokenSigner`, `authMiddleware`, RBAC roles `admin/operator/viewer/device`). No mTLS in the built server.
- Device endpoints: `GET /client/update` (204/200) and `POST /client/telemetry`; device acts only on its own token subject.

Canonical corpus ground truth: `endpoints.md` (rev 2), `schema.md`, `architecture.md`, master design. These match the built server closely (Gin, releases/deployments, `all-targets` MVP, opaque IDs, SHA-256+signature+AVB, staged rollout → 1.0.1, TUF → 1.0.1+).

§2. document inventory

Base: `additions/initial_research_03/helix_ota_big/`. 26 `.md` files.
`Binaries/exports` (`.pdf/.docx/.html/.tar/.zip`) intentionally not opened (duplicate renderings of the `.md`).

#	Document (path)	Lines	What it specifies
D0	<code>CODE_REVIEW_AND_GAP_ANALYSIS.md</code>	539	Independent review of the addition's OWN c CRITICAL + 9 HIGH + 8 MEDIUM + 7 LO are internal contradictions (endpoint paths, e convention, validation order, Go module pat the addition's own files. Reconciled in §5.
D1	<code>README.md</code>	153	Project overview, version roadmap matrix, d vasic-digital submodule list, 8 new <code>helix-*</code> Module org: <code>HelixDevelopment</code> + <code>vasic-o</code> RSA-4096 + SHA-256 + A/B; plugin OS-ad

#	Document (path)	Lines	What it specifies
D2	1.0.0-mvp/server/ SERVER_IMPLEMENTATION.md	3968	Full Go server design: project layout (chi router, internal/handler service repository), go.mod (github.com/vasicek/helix-ota-server, chi, pgx, RSA), services (UpdateService/DeviceService/RolloutService/ArtifactService/TelemetryService/AuthService), decision engine (DRAFT→RUNNING→PAUSED→HALTED→COMPLETED, deterministic SHA-256 device bucketing, dwell-time advance, auto-liveness (failure-rate), validation pipeline (Structure→Hash→Signature→Compatibility), SHA-256, RSA-PSS verify, worker pool), metrics, enrollment, telemetry, background jobs, middleware, main, submodule integration.
D3	1.0.0-mvp/client/android/ ANDROID_CLIENT_DESIGN.md	3202	Android 15 / RK3588 client: Virtual A/B (SecureBoot), update_engine Binder IPC + ApplyPayload, partition layout, OTA zip/payload.bin/payload_properties.txt/care_map.pb, resumable Range download, SHA-256+RSA verification, engine, AOSP build integration (/system/property), UX, Go client SDK (gomobile AAR).
D4	1.0.0-mvp/docs/api/ REST_API_SPECIFICATION.md	3333	REST surface: auth (login/refresh/logout/TOTP), device mgmt (/devices/register), artifact mgmt (/artifacts/upload, /artifacts/{id}/download), update check (POST /updates/check), rollouts (rollouts), telemetry/reporting, Dashboard V1 (underspecified), Go struct appendix, error/rollback appendices (clipped).
D5	1.0.0-mvp/docs/ VERSION_ROADMAP.md	800	Version contract MVP→Rollback→Delta→Linux→Windows. MVP feature lists (server/client/dashboard) + criteria. Conflicting endpoint paths (POST /devices/register, GET /devices/{id}/update-check, POST /artifacts).
D6	1.0.0-mvp/docs/architecture/ SYSTEM_ARCHITECTURE.md	2795	Full system architecture; Go structs (DeviceService, NotificationService, DashboardSession, validation chain (Hash-fingerprint)). Conflicts with DB schema on field names (payload vs. update).
D7	1.0.0-mvp/docs/database/ DATABASE_SCHEMA.md	1691	PostgreSQL 16 DDL: UUID v4 PKs, enums (rollout_status, device_status, upload_status, rollout_status), CHECK, tables (devices, device_groups, artifacts, updates, rollouts, rollout_stages, telemetry_payloads, users, api_keys, audit), indexes, partitioning, strategy.
D8	1.0.0-mvp/docs/security/ SECURITY_ARCHITECTURE.md	1908	STRIDE threat model; transport (mTLS), artifact verification (SHA-256 + RSA + chain), authn/authz (JWT + API keys + TOTP), device identity/provisioning, update-safety guarantees, server compliance/audit, security testing.
D9		2351	

#	Document (path)	Lines	What it specifies
	1.0.0-mvp/docs/submodules/ NEW_SUBMODULE_SPECIFICATIONS.md		Specs for 8 new helix-* submodules (server, android, dashboard, update-engine, artifact-validation, rollout-engine, device-identity). Module path dev.helix.ota.* (conflicts with D2). Contains integration, release checklist, dependency management.
D10	1.0.0-mvp/docs/testing/ TESTING_STRATEGY.md	2043	Four-layer fix verification, mutation testing (integration/e2e/HIL, anti-bluff; CI/CD section).
D11	1.0.0-mvp/docs/deployment/ DEPLOYMENT_GUIDE.md	1996	Docker/K8s deployment; env vars; monitoring (referenced, partly missing).
D12	1.0.0-mvp/docs/research/ OTA_SYSTEMS_RESEARCH.md	1964	Survey of existing OTA solutions (overlaps with research/ota_landscape_report.md + research/ota_stacks/).
D13	1.0.0-mvp/containers/docker- compose.yml	793	Full stack: Traefik (TLS/ACME/mTLS forwarding), helix-ota-server (chi per comment), dashboard (nginx), PostgreSQL 16 (tuned), Redis 7, Minio (versioning), HashiCorp Vault (transit RSA-signing, kv-v2, TOTP), minio-init, vault-init, vault-secrets).
D14	1.0.1-rollback/docs/ ROLLBACK_DESIGN.md	2127	1.0.1 rollback: auto (boot-failure) / server-triggered; user-initiated; rollback vs downgrade; versioning; mgmt; data safety; rollback API; Go + Kotlin impls.
D15	1.0.2-delta-updates/docs/ DELTA_UPDATES_DESIGN.md	1969	1.0.2 delta/differential OTA; AOSP block diff; delta-gen submodule; delta selection; 60-90% bandwidth reduction; DB+API additions; better error handling.
D16	1.1.0-linux-support/docs/ LINUX_OTA_RESEARCH.md	2373	1.1.0 Linux OTA: apt/rpm-ostree/pacman/genfs; rootfs; OSAdapter plugin interface; new submodule migration path.
D17	1.2.0-windows-support/docs/ WINDOWS_OTA_RESEARCH.md	2909	1.2.0 Windows OTA: WUA/WSUS/WUfB/Windows Update; Windows Service client; MSI/MSIX; Windows Defender (code signing); Windows adapter; submodule migration path.
D18	2.0.0-multi-os-universal/docs/ MULTI_OS_UNIVERSAL_DESIGN.md	1466	2.0.0 universal: dynamically-loaded OS-adaptation system; OS catalog; OS-agnostic pipeline; cross-platform dashboard + dependency coordination; plugin architecture.
D19– D26	1.0.0-mvp/docs/diagrams/*.md (8 files)	79– 258 ea	Mermaid: high-level arch, update flow, rollout flow, client state machine, artifact validation, data flow, security layers, container arch. Overlap canonical master/diagrams/.

§3. requirement inventory table

Coverage is vs the union of the **canonical corpus** and the **built server**. “Built” = present in server/. “Corpus” = present in a spec doc. Source §s reference the addition.

ReqID	Requirement (source)	Category	Coverage
R01	Device registration endpoint	API	COVERED — built POST /devices/register Addition self-conflicts on path (D0/C1).

ReqID	Requirement (source)	Category	Coverage
R02	Device registration body fields	API/Model	PARTIAL — built DeviceRegistration{hardware_id,model_id}(wire.go:41). Addition wants slot_suffix
R03	Update check (device-facing)	API	COVERED — built GET /client/update §12.1. Addition self-conflicts GET vs POST
R04	Artifact upload + validation	API	COVERED — built POST /artifacts/upload §9.1, artifact_validation.md.
R05	Artifact download w/ Range, identity encoding	API	COVERED — corpus endpoints.md §3/ returns reference URL (artifactURL). Byt
R06	Release create/list/get	API	COVERED — built /releases CRUD (se no “release” concept (uses “update”).
R07	Deployment (all-targets) create/get	API	COVERED — built /deployments (handl
R08	Staged/phased rollout engine (5/10/30/50/100, dwell, auto-halt)	Rollout	PARTIAL/DEFERRED — corpus defers to engine.InCohort. Addition gives a full G
R09	Deterministic device→cohort hashing	Rollout	COVERED — built rollout.InCohort s §4.3) is a compatible reference.
R10	Telemetry ingest (batch)	API	COVERED — built POST /client/telem §12.2, telemetry_processing.md.
R11	Auth: login/refresh + JWT + RBAC	Security	COVERED — built /auth/login,/auth/
R12	Artifact integrity: SHA-256 + signature, server+device	Security	COVERED — built ed25519 detached sig
R13			

ReqID	Requirement (source)	Category	Coverage
	A/B + AVB/dm-verity + auto boot-failure rollback	Client/ Safety	COVERED — corpus master §1, client_stacks/android-avb-rollback.md. Add
R14	Virtual A/B + payload.bin + applyPayload headers	Client	COVERED — corpus research/stacks/update_engine_integration.md. D3 ad
R15	Validation chain order	Security/ Safety	COVERED (different order) — corpus/build Hash→Signature→Structure→Compati
R16	Anti-rollback / downgrade prevention	Safety	MISSING (MVP-critical) — addition SA2 server+client check. Built handleClientU NOT forbid a release whose version < device on release creation only.
R17	One active rollout/deployment per group	Rollout	PARTIAL — built rejects overlapping active → 409, handlers_deployment.go:65). Addit
R18	Mandatory update enforcement behavior	API/Client	MISSING — addition M7: mandatory/de (mandatory field) nor corpus.
R19	User management CRUD	API	MISSING — addition H1; RBAC matrix e UserDirectory only (server.go:22). Corpu
R20	Device-group management CRUD	API	MISSING — addition H2. Built carries gr deployments accept optional group.
R21	API-key management CRUD	Security/ API	MISSING — addition H3; D8 defines key
R22	Dashboard WebSocket/SSE wire protocol	API	MISSING — addition H4 (under-specified
R23	Dashboard Web UI design doc	Dashboard	MISSING — addition H5; no dashboard d yet.
R24	Go Client SDK design doc	Client	PARTIAL — addition H6 wants full SDK c integration_guide.md+build_integra undocumented.
R25	Rollout/Device model [?] DB alignment	Model	PARTIAL — addition H7/H8 are internal a yet). Canonical schema.md is the target.
R26	Telemetry event endpoint (rich events) distinct from device status	API	PARTIAL — addition H9. Built telemetry I EventDownloadStarted/Installing/Success/I events) + /telemetry/overview not bui
R27	Tampered-artifact server-side alerting (SECURITY_TAMPER_DETECTED)	Security	MISSING — addition SA3. Built: device I failure-rate only.
R28	mTLS device transport	Security	NOT-BUILT — addition mandates mTLS (device JWT bearer ; corpus transport_s

ReqID	Requirement (source)	Category	Coverage
R29	Vault for signing/secrets	Deployment/ Security	NOT-BUILT — addition compose runs Vault ed25519. Corpus <code>key_management.md</code> = s
R30	Traefik ingress / TLS / ACME	Deployment	PARTIAL — addition compose uses Traefik deployment/overview.md.
R31	Redis cache/session/ratelimit	Infra	OPTIONAL — addition compose requires Redis only if needed.
R32	PostgreSQL schema (devices/ artifacts/releases/rollouts/ telemetry/users/groups/ api_keys/audit)	DB	PARTIAL — canonical <code>schema.md</code> exists; reference (partitioning, indexes, enums, au
R33	Mutation testing $\geq 85\%$, four-layer verification	Testing	COVERED — corpus tests/test_strat
R34	Containerized via containers submodule	Deployment	COVERED — master §3; addition compos
R35	8 new <code>helix-*</code> submodules	Submodules	PARTIAL/DIVERGENT — addition D9 na dashboard/-update-engine/-artifact repos are ota-protocol/ota-artifact-engine under HelixDevelopment.
R36	OS-adapter seam (universality)	Architecture	COVERED — master §4 OS-adapter seam

§4. conflicts with locked decisions and the built server

#	Addition position	Locked / built reality	Resolution
X1	chi router (go-chi/chi/v5, <code>SERVER_IMPL</code> §1–2; compose comment “Chi router”)	LOCKED Gin (D6); built uses <code>Gin</code> (<code>server.go</code>).	Gin wins. Discard <code>chi</code> . The addition’s handler/service/repository layering is still a useful reference for internal seams (ADR-0003 modular monolith).

#	Addition position	Locked / built reality	Resolution
X2	RSA-4096 / RSA-PSS artifact signing (SERVER_IMPL §5.4; README; compose Vault type=rsa-4096)	Built ed25519 detached signature + SHA-256 (server.go, ota-artifact-validator); corpus signing_verification.md.	ed25519 wins (built + canonical). RSA-4096 is heavier and not what's implemented. If RSA support is ever desired it must be an additive, justified ADR change — not a silent swap.
X3	mTLS device transport + Device CA provisioning (SECURITY §3/ §6; compose device-mtls router)	Built device JWT bearer ; corpus transport_security.md (HTTP/3→HTTP/2 + bearer).	JWT-bearer wins for MVP. mTLS is a valid hardening path; record as optional in security/ but do not block MVP.
X4	Module path github.com/vasic-digital/helix-ota-server (D2) and dev.helix.ota.* (D9)	Built github.com/HelixDevelopment/helix_ota/server; deps under HelixDevelopment/ota-.*.	Built path wins. Addition is internally inconsistent (its own C10).
X5	Domain vocabulary updates + rollouts at MVP (D4/ D6/D7)	Built/canonical releases + deployments ; staged rollouts deferred to 1.0.1.	releases/ deployments wins. Addition Update→Release, addition Rollout→ (deployment now / staged-rollout 1.0.1).
X6	Full staged-rollout decision engine in MVP (SERVER_IMPL §4)	Locked scope: MVP = all-targets; staged → 1.0.1 (master §1 non-goals; built strategyAllTargets).	Defer. Keep the engine algorithm as a 1.0.1 design seed (high-quality reference).
X7	Vault as signing/ secret store (compose)	Corpus: security brick + container secrets first (§11.4.74); synthesis C7-adjacent.	Catalogue-first. Evaluate security brick before adopting Vault.
X8	UUID-v4 PKs exposed; OR prefixed string IDs (D7 vs D5/D6 — addition's own C11)	Built/canonical: opaque server-issued strings , UUID-formatted in OpenAPI, treated opaque by clients (endpoints.md §2, newRandomID).	Opaque-string wins. Internal UUID vs external-id split (addition's own recommendation) already satisfied by

#	Addition position	Locked / built reality	Resolution
			the opaque-id convention.
X9	Coverage target 85% (D5/D10)	Synthesis C8: floor ≥90% for safety-critical paths (signing/apply/rollout-gate).	≥90% floor for safety paths; 85% elsewhere acceptable.

No conflict was found with: A/B+AVB+auto-rollback, SHA-256 integrity, staged percentages (5/10/30/50/100), telemetry event stream, PostgreSQL+MinIO, containerization, REST-primary, mutation/four-layer testing — these AGREE with locked decisions and are high-confidence inputs.

§5. reconciling the addition's code_review against the actual build

The addition's CODE_REVIEW_AND_GAP_ANALYSIS.md (D0) audits the **addition's own documents against each other**. It does NOT see the built server. Re-scored against the real build + canonical corpus:

CRITICALs that are NOT real against the build (already resolved by canonical choices):

- **C1 (device-register path), C3 (update-check method/path), C4 (artifact-upload path):** The built server already fixed canonical paths — POST /devices/register, GET /client/update, POST /artifacts/upload. The addition's contradictions are artifacts of its own multi-author generation; not blockers for Helix.
- **C5/C6/C8 (rollout/device/validation status enums diverge across addition docs):** Built uses typed enums from ota-protocol/ota-telemetry-schema (single source). No three-way enum drift in the real system. Real residual: ensure wire enum strings are documented (mostly done in endpoints.md rev 2 — telemetry 6-value enum vs 7-value update-state).
- **C7 (os_type CHECK vs API enum):** Built has no DB yet; defaultAndroidPolicy accepts Android only (Phase-1). When pgx lands, use application-level validation (not a brittle CHECK) — matches addition's own recommendation.
- **C10 (Go module path):** Built path is canonical (HelixDevelopment/helix_ota). Addition's two-path conflict is moot.
- **C11 (UUID vs prefixed string IDs):** Resolved by the opaque-id convention (X8).
- **C2 (registration body schema drift):** Built body is canonical (DeviceRegistration). Optional slot_suffix/hardware_fingerprint are the only real takeaways (R02).

Findings that ARE real gaps against the current build (MVP-relevant first):

- **SA2 / “anti-rollback protection” (→ R16): REAL, MVP-critical.** No explicit `target_version > device.current_version` downgrade block exists in the built update-check or release validation. Steady-state ≥ 204 is not a downgrade guard. **Add server-side + (spec) client-side downgrade prevention.**

- **C9 / SA1 “validation chain order” (→ R15): REAL but indirect.** Built validation lives in ota-artifact-validator; the hash-before-signature ordering must be **verified** there and documented. (UNVERIFIED current order in the external module.)
- **M6 “concurrent rollout overlap” (→ R17): PARTIALLY real.** Built already 409s on overlapping active deployment per target set; the DB partial-unique index is a future-pgx hardening, not a current bug.
- **M7 “mandatory update behavior” (→ R18): REAL gap** (undefined everywhere).
- **H1/H2/H3 (user / device-group / api-key CRUD → R19/R20/R21): REAL gaps** (not built, not in corpus). Severity is post-MVP unless the dashboard needs them.
- **H4 (WebSocket protocol → R22), H5 (dashboard design → R23), H6/L7 (SDK doc + gomobile constraints → R24): REAL doc gaps** (dashboard phase not yet authored).
- **H9 (telemetry history/overview reads → R26): PARTIALLY real** — ingest is built; read/aggregate endpoints are not.
- **SA3 (tamper-detected event → R27): REAL gap.**
- **M1 (download URL signing): REAL but a documented design choice** — built returns a reference URL; pre-signed vs direct-auth must be decided in endpoints.md/Storage-brick (UNVERIFIED which the Storage brick provides).
- **M2–M5, L1–L6 (provisioning guide, AOSP integration guide, CI/CD doc, migration strategy, runbooks, .env.example, SLOs, backup/restore): REAL doc gaps**, low priority; route to ops/security/infra docs as those phases land.

Net: Most of the addition’s 11 CRITICALs are self-referential and already settled by the canonical build. The genuinely actionable, build-relevant items are **R16 (anti-rollback — MVP-critical)**, **R18 (mandatory-update)**, **R15 (verify validation order)**, **R27 (tamper event)**, **R26 (telemetry reads)**, and the **CRUD/doc gaps R19–R24**.

§6. future-phase catalogue

Each future-phase doc maps to an existing versioned dir under docs/research/main_specs/. The addition uses 1.0.1-rollback / 1.0.2-delta-updates / 1.1.0-linux-support / 1.2.0-windows-support / 2.0.0-multi-os-universal; the canonical corpus currently has 1.0.1-staged-rollout, 1.X-linux, 1.X-windows, 1.X-other-os. Note the **numbering divergence**: addition slots rollback at 1.0.1 and delta at 1.0.2, whereas canonical reserves **1.0.1 for staged-rollout** (locked: staged rollout is the first post-MVP increment). Recommendation: keep canonical 1.0.1 = staged-rollout; place addition’s rollback/delta material as **1.0.2 / 1.0.3** (or sub-sections) so the staged-rollout engine — which both depend on — lands first.

Addition doc	Material	Destination dir	Notes
D14 ROLLBACK_DESIGN (1.0.1)	server-triggered + user-initiated + multi-version	1.0.X-rollback/ (new; suggest 1.0.2 after staged-rollout)	Auto boot-failure rollback is ALREADY MVP (A/

Addition doc	Material	Destination dir	Notes
	rollback, version history, rollback API, Go/ Kotlin impls		B+AVB). This doc covers the <i>added</i> rollback types — matches locked “end-user/multi-version rollback deferred past MVP”.
D15 DELTA_UPDATES (1.0.2)	AOSP block-diff deltas, helix-delta-gen, delta selection, DB+API additions, benchmarks	1.0.X-delta-updates/ (suggest 1.0.3)	Aligns with ADR-0005 (delta updates). Fold benchmarks as UNVERIFIED until measured.
D16 LINUX_OTA_RESEARCH (1.1.0)	apt/rpm-ostree/ pacman/ generic A/B; OSAdapter interface; migration path	1.X-linux/ (exists)	Strong research; reconcile OSAdapter interface with master §4 OS-adapter seam.
D17 WINDOWS_OTA_RESEARCH (1.2.0)	WUA/ WSUS/ WUfB/ Intune; Windows Service client; MSI/ MSIX; code-signing	1.X-windows/ (exists)	“augment not replace Windows Update” thesis.
D18 MULTI_OS_UNIVERSAL (2.0.0)	dynamic OS-adapter plugin system; cross-OS dashboard + dependency coordination; plugin manifest	1.X-other-os/ (exists) or new 2.0.0-multi-os-universal/	Plugin/manifest design = the universality endgame; reconcile with OS-adapter seam.
D12 OTA_SYSTEMS_RESEARCH	OTA solution survey	merge into research/ ota_landscape_report.md + research/stacks/	Likely redundant with existing landscape

Addition doc	Material	Destination dir	Notes
			report; diff before folding.
D19–D26 diagrams	Mermaid arch/flow/ state/ER/ security/ container	merge into 00-master/ diagrams/	De-dup against existing diagrams.

§7. folding actions summary

MVP-critical (do for 1.0.0): 1. **R16 anti-rollback / downgrade prevention** — add `target_version > device.current_version` enforcement server-side (update-check) + release validation; document in `endpoints.md` §12.1 and `artifact_validation.md`. (Addition SA2.) 2. **R15 validation-order verification** — verify `ota-artifact-validator` performs hash-before-signature; document the order (currently UNVERIFIED). (Addition C9/SA1.)

Near-term / 1.0.1 candidates: 3. **R18 mandatory-update behavior** (M7) — define mandatory/deadline semantics. 4. **R27 tamper event** (SA3) — `SECURITY_TAMPER_DETECTED` halts rollout + alerts; into `1.0.1-staged-rollout/` + `00-master/threat_model.md`. 5. **R08 staged-rollout engine** — fold addition’s decision-engine algorithm (dwell, auto-halt, deterministic bucketing) into `1.0.1-staged-rollout/` as the design seed. 6. **R17 one-active-per-group DB index** when `pgx` lands.

Doc/phase work (route, don’t build now): 7. R19–R24 (user/group/api-key CRUD, WebSocket protocol, dashboard design, SDK+gomobile doc) → respective phase docs. 8. R32 — mine D7 (`DATABASE_SCHEMA`) for partitioning/index/audit detail when authoring the `pgx` implementation; reconcile enums to canonical wire strings. 9. Future-phase docs D14–D18 → versioned dirs per §6 (mind the 1.0.1 numbering divergence: staged-rollout stays 1.0.1).

Reject (conflict with locked/built): `chi` (X1), `RSA-4096 swap` (X2), `mTLS-only MVP transport` (X3), `vasic-digital/dev.helix.ota.*` module paths (X4), `updates/rollouts MVP vocabulary` (X5), `Vault-first secrets` (X7), `exposed-UUID PKs` (X8). Reuse their surrounding design ideas where additive and justified.

End of addition #3 analysis.